SUMMARY

- A device used to study the conduction of electricity through partical gases at low pressure is known as gas discharge tube
- Charge per unit mass is known as specific charge or normalized charge, and it is independent of gas used and also nature of electrodes used in the experiment.
- The minimum distance between alpha particle and centre of the nucleus just before it gets reflected back by 180° is defined as distance of closest approach r_0.
- The impact parameter (b) (see Figure 9.12) is defined as the perpendicular distance between the centre of the gold nucleus and the direction of velocity vector of alpha particle when it is at a large distance.
- According to Bohr atom model, angular momentum is quantized.
- The radius of the orbit in Bohr atom model is $r_n = a_0 \frac{n^2}{Z}$
- The radius of first orbit hydrogen atom is $a_0 = \frac{\varepsilon_0 h^2}{\pi m e^2} = 0.529\,\text{Å}$ and it is also known as Bohr radius
- The velocity of electron in n^{th} orbit is $\upsilon_n = \frac{h}{2\pi m a_0}\frac{Z}{n}$ ms^{-1}.
- The fine structure constant is $\alpha = \frac{1}{137}$ which is a dimensionless constant
- The total energy of electron in the nth orbit is $E_n = -\frac{me^4}{8\varepsilon_0^2 h^2}\frac{Z^2}{n^2} = -13.6\frac{Z^2}{n^2} eV$.
- The energy required to excite an electron from the lower energy state to any higher energy state is known as excitation energy and corresponding potential supplied is known as excitation potential.
- The minimum energy required to remove an electron from an atom in the ground state is known as binding energy or ionization energy.
- The potential difference through which an electron should be accelerated to get ionization energy is known as ionization potential.
- The wavelength of spectral lines of Lyman series lies in ultra-violet region
- The wavelength of spectral lines of Balmer series lies in visible region while those of Paschen and Brackett series lie in infra-red region
- The nucleus of element X having atomic number Z and mass number A is represented by ${}^{A}_{Z}X$.
- The emprical relation connecting radius of nucleus ($Z > 10$) R and mass number A is given by $R = R_0 A^{1/3}$ where $R_0 = 1.2\,F$
- The density of nucleus $\rho = 2.3 \times 10^{17}\, kg\, m^{-3}$, and the nuclear matter in a highly compressed state.
- If M, m_p and m_n are masses of a nucleus (${}^{A}_{Z}X$), proton and neutron respectively , then the mass defect is $\Delta m = \left(Zm_p + Nm_n\right) - M$

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